Group ADJ Midterm Report

Yuzhou Zhou
Zhean Zhang
Andaman Khunaprapakorn
Data Science Institute, Brown University
https://github.com/jimmy0303/data1030-wildfire-project

1. Introduction

In response to a wildfire, first responders must decide what resources to allocate to combat which fire (i.e., first-response crews, helicopters). To make this decision, they must go off of whatever limited information they have at that given time. When factoring in how wildfire sizes tend to vary widely, with 88% of fires staying under 10 acres and 5% exceeding 100, it is important to decide which wildfires have the highest likelihood to become "large" and allocate resources accordingly in the case where there are multiple simultaneous fires.

Hence, the goal of this project is to make a model to predict the severity (peak size) of wildfires using information available during the fire's early stages.

Specifically, the aim of the project is to classify wildfires into four ordinal size classes (defined in Section 2).

To prevent data leakage, only features available at a fire's initial reporting are used to predict our target variable (final size/class)

1.1 Previous work

Many previous projects in this space are based on the Fire Program Analysis Fire-Occurrence Database (USDA Forest Service), which is a government-owned US wildfire database. These projects tend to predict based on location, date, and weather. Although these projects typically have a high final accuracy, a deeper look into these projects revealed that these scores are highly skewed by correctly predicting the smallest class, which masks the limited accuracy for predicting class D+ wildfires. (Vijay)

Together, other works cover a broader geographical range, such as Coffield et al., which explores wildfire Machine Learning classification in Alaska. In this paper, they accomplished a 50.4 ± 5.2% accuracy for 3 classes, which were classified as small, medium, and large. This 50.4% accuracy figure shows that the classification task at hand is highly complex and "unsolved".

2. Data

Our wildfire records came from the NIFC WFIGS Incident Locations API. We queried eleven western U.S. states for 2020–2025 UTC discovery dates. Each record needed a reported incident size, an out date, and initial coordinates. Records were downloaded in batches and saved in CSV format.

The download contained 75,404 records. We dropped 132 records with coordinates outside 31–50°N and 125–102°W, leaving 75,272 unique incidents. The 33 columns include numeric, categorical, date, ID, and quality fields. These filters leave out some fires.

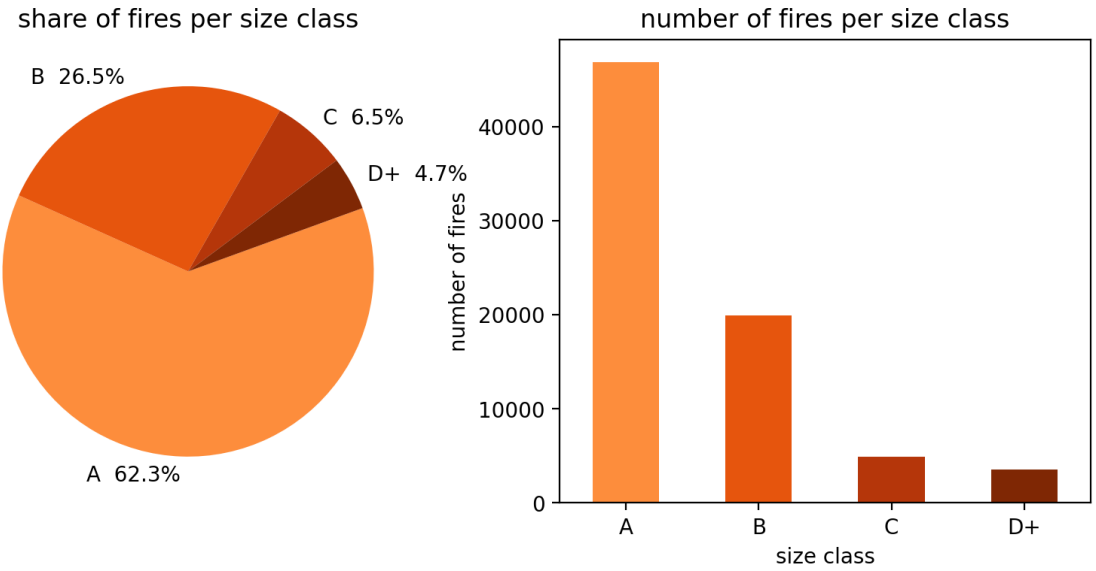
We used reported IncidentSize in acres to define the ordered target, size_class. Class A covers 0–0.25 acres, B covers more than 0.25 but less than 10, C covers 10 to less than 100, and D+ covers 100 or more. These classes contain 46,889, 19,949, 4,907, and 3,527 incidents, respectively. Reported size may differ from final burned area.

We used coordinates rounded to 0.5° for weather requests. We obtained ERA5 weather from the Copernicus Climate Change Service/ECMWF through Open-Meteo's Historical Weather API. Each fire was matched to its request location and the seven complete UTC days before discovery. Using pandas, we created ten weather features from temperature, precipitation, wind, humidity, reference evapotranspiration, and vapour pressure deficit, including dry-day counts. Elevation came from the Copernicus GLO-90 DEM through Open-Meteo's Elevation API.

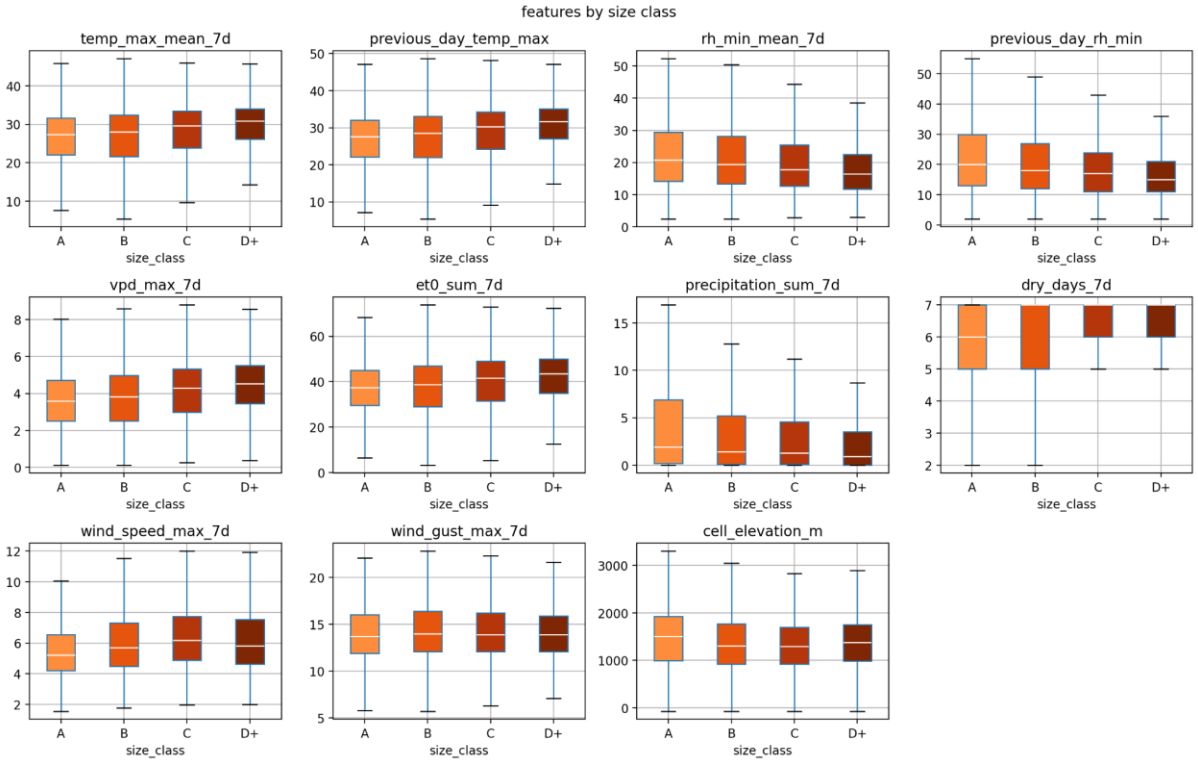
We selected candidate variables to describe recent weather and differences in location and season. We have not tested their predictive value. Weather and elevation refer to rounded locations rather than exact fire sites. incident_size_acres and final_acres describe the outcome, so we must exclude them as predictors.

Each retained incident has complete seven-day weather and elevation data. Fuel group is missing for 54,956 incidents, landowner category for 556, and FinalAcres for 74,222. We left these missing values unchanged. We will drop any predictor not recorded at discovery. ERA5 describes past weather, but earlier dates do not show that the data were available at discovery.

3. Exploratory Data Analysis (EDA)



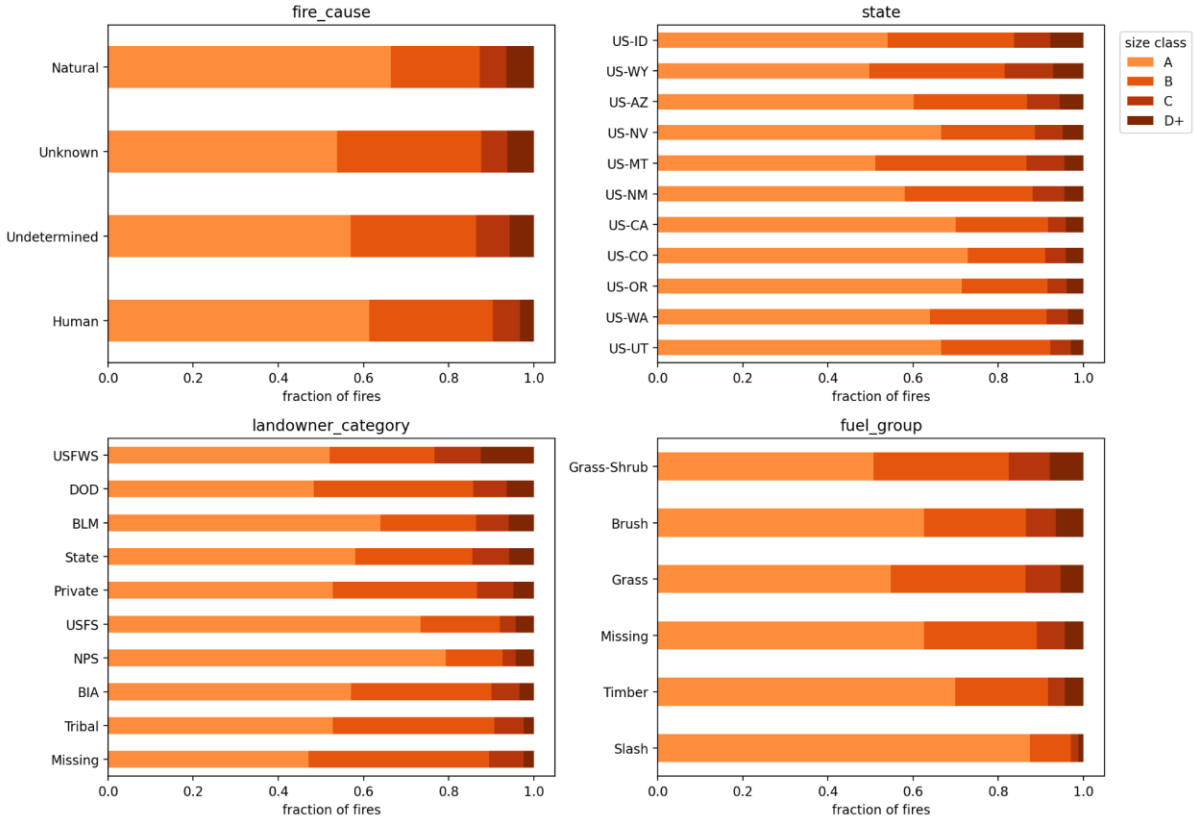
After the previous data processing, we first checked whether our target classes are balanced, then explored how each feature relates to the fire size, and finally looked at the time-related variables. We found that our sample size is not balanced enough compared with our expectation. According to this picture, we can see that the more serious fires (C and D+) only contain 11.2%. In this situation, we are going to use F1-score rather than accuracy as our main evaluation metric and also set the class_weight to reduce the impact created by the imbalanced sample.



(outliers were hidden in order to have better readability)

These graphs show how the weather in the week before is related to the fire size. From this set of graphs, we could find out that for more serious fires to appear, the weather a week before will be drier and hotter. The wind gust and the elevation provided a relatively weak

connection with the fire size. By looking through these box plots, we could see that there are lots of overlaps, which means a single feature might not be useful for our model. We might need combinations of different features to improve our model performance in the future.



These stacked bar plots show how the size class changes across different categorical features. According to these graphs, we could see that fires caused by natural reasons are more likely to become serious than human-caused fires. Fires in Wyoming and Idaho, and fires in grass–shrub fuel, also tend to be larger, and the slash fuel is mostly linked to small fires. Thus, the fire cause and location could be useful features for our model.



We also considered whether there is a seasonality/time pattern in fires. In the first plot, most fires happen in summer(June to September). The second plot shows that the share of large fires is also higher in July and August, while spring months have more medium-sized fires. The last plot shows that the fire size balance stays fairly stable across years, which means splitting the data by time should not change the target distribution too much.

4. Problem description and potential impact

4.1 Problem Description

In a scenario with multiple simultaneous wildfires, first responders must allocate resources based on the latent severity of a fire to mitigate the overall impact. To do so effectively, they must be able to determine the latent "danger" of a fire, quantified by its size.

Hence, our task addresses this as a supervised ordinal classification problem and aims to predict the target variable (final size/category of wildfires) using the conditions at initial reporting. Features that will be used in the prediction include:

1. Weather, including a 7-day history leading up to the incident
2. Season
3. Elevation
4. Location
5. Others: Ignition cause, Land-use category, fuel group.

Each prediction sorts wildfires into the four size classes defined in Section 2.

Each prediction is made once and will be done so only with features available in a real-world crisis response.

We will evaluate the success of our project based on the F-1 score, mainly of our final predictions.

4.2 Potential Impact

The main use case for this model is in the aforementioned dispatch scenario, which will aid the distribution of resources such as first-response teams, aircraft, and rescue vehicles.

If accurate enough to deploy into the real world, the model also allows for the forecast of the danger-zone of each fire, informing local authorities of which areas need to be evacuated. Other benefits also include: potential forecasts of infrastructure damage, danger-level to firefighters, or a home's risk to fire given its location.

4.3 Risks and Ethics

Even if there is evidence of the model's performance, key risk and ethical concerns should be raised before any real world implementation is considered. Firstly, the data is trained on historical data. Hence, by the time of implementation, there is a real likelihood that the model is "outdated", and is not properly tuned to reliably predict fire sizes. Secondly, the model's predictions should not be treated as fact; doing so would undoubtedly lead to cases of break-out large fires the model did not catch, an ethical hazard which may lead to real civilian casualties.

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